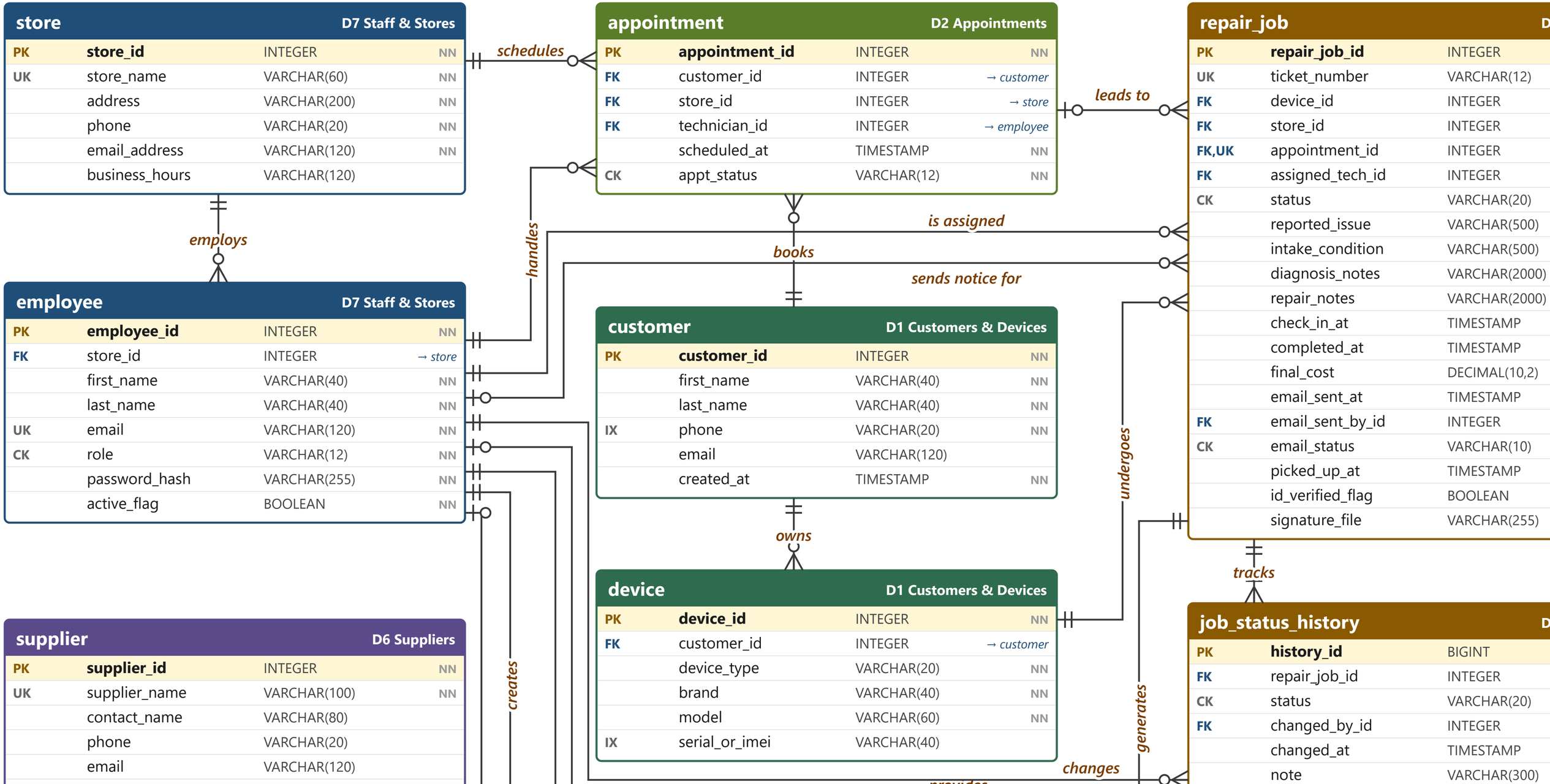
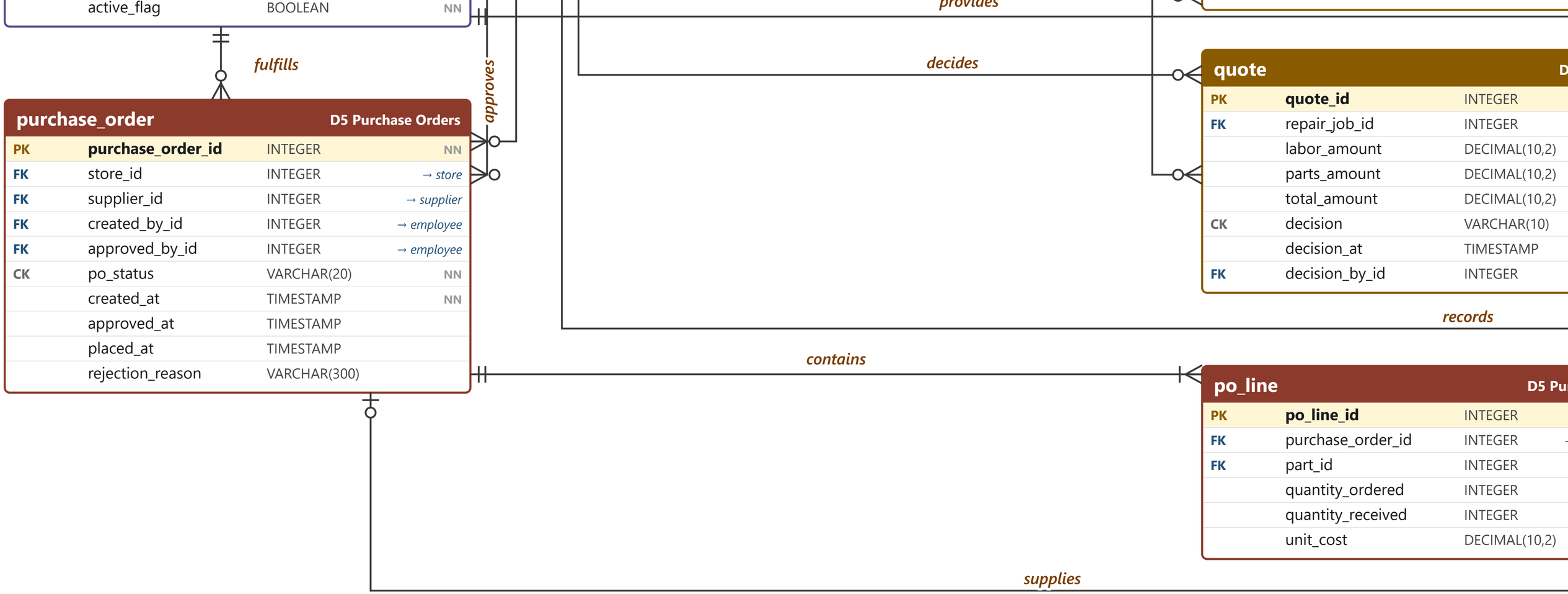
QuickPhone — Physical ERD (complete): 16 tables with all keys and attributes

One diagram for the whole data model. Each verb reads from the "one" side to the "many" side. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null · derived = denormalized





Every foreign key is drawn as a relationship line. EMPLOYEE plays several roles: the technician assigned a job, the sender of the completion notice, and the person who changes a status, decides a quote, records a stock change, or creates or approves a purchase order.

QuickPhone Repair Tracking System: Physical ERD notes

MIS 4173 Sprint 3 · Team 4 · Method: Dennis, Wixom, Roth & Haley, *Systems Analysis and Design*, 8th ed., Ch. 5 and Ch. 10

Logical to physical (Fig. 10-7)

1. **Entities to tables.** Sixteen tables in snake_case; each header names the DFD data store (D1–D7) that holds it, so the ERD and the DFDs stay balanced.
2. **Attributes to fields.** Each field has a data type and size (VARCHAR, INTEGER, DECIMAL, TIMESTAMP, BOOLEAN), NN for not null, and CK where only listed values are allowed (statuses, roles, payment method).
3. **Primary keys.** Every table has one (PK); store_part_stock uses the pair store_id + part_id.
4. **Foreign keys.** Every relationship is a foreign key in the child table (FK, with the parent table named). Every foreign key is drawn as a line with crow's-foot cardinality, and each line carries one active verb read from the "one" side to the "many" side (Ch. 5).
5. **System-related components.** created_at and other time stamps, password_hash, id_verified_flag, signature_file, and the audit tables job_status_history and inventory_txn.

Documented denormalizations

The logical ERD is in third normal form. Three fields marked *derived* are stored on purpose for speed and are kept correct by the program (Ch. 10, optimizing storage):

- `repair_job.status`: the latest row of job_status_history.
- `quote.total_amount`: labor_amount + parts_amount.
- `store_part_stock.quantity_on_hand`: the sum of inventory_txn.quantity_change.

Assumptions

Types are generic SQL; no database product is chosen yet. EMPLOYEE.store_id is a home store; the Owner sees both stores. Photo ID numbers are never stored, only that the check was done.